PCB Defect Detection Under Image Misalignment: Data Report

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TABLE I
PAIRS AND BOXES PER SPLIT AFTER THE GROUPED PARTITION.

Split	Pairs	Open	Short	M.bite	Spur	Copper	P.hole	Boxes
Train	800	1005	825	1032	888	778	810	5338
Val	200	278	203	347	254	232	221	1535
Test	500	659	478	586	483	464	470	3140
Total	1500	1942	1506	1965	1625	1474	1501	10013

***Abstract*—This report describes the data used to test whether a compact learned detector can separate genuine PCB defects from misalignment artefacts. It covers the DeepPCB source, the grouped train/validation/test partition, a quality audit of the raw pairs, the seeded misalignment protocol, and the checks applied to every prepared image and box. All numbers are produced by the scripts and JSON reports in the project repository.**

I. SOURCE DATASET

DeepPCB [1] contains 1,500 pairs of 640×640 binarised images: a defect-free *template* (reference) and an *inspected* image of the same PCB region, aligned by the authors with template matching. Each pair has 1–15 axis-aligned boxes (mean 6.7, 10,013 in total) labelled with one of six classes (Table I). Defects are small: median box size 35×33 px (18–173 px wide), about 0.3 % of the image area. Classes are close to balanced (largest/smallest ratio 1.33). Most defects were added artificially by the dataset authors, which limits claims about real production defects. The official lists keep 1,000 *trainval* and 500 *test* pairs from 11 source-board *groups* (folder names); group is the only board identifier available.

II. PARTITION

The official 1,000/500 split is kept. About 20 % of trainval is held out for validation *before* any augmentation, grouped by source board so that no board appears in both train and validation. Every subset of trainval groups containing 17–23 % of the pairs was scored by the L_1 distance between its class proportions and those of all trainval; the best is group `00041` (200 pairs, $L_1 = 0.059$), giving **800 train / 200 val / 500 test**. All shifted variants of a pair inherit the pair's split.

Because the official test list was kept, the proposal's rule "same board, same partition" cannot hold for test: 7 of the 11 test groups also occur in trainval (168 test pairs), including 21 from the validation group. The other 332 test pairs come from four groups never seen in trainval (`12000`, `12100`, `12300`, `90100`), so a group-held-out validation set is a reasonable proxy for test.

III. DATA-QUALITY AUDIT

Format. All 3,000 images are 640×640; 62 images in 57 pairs are stored as 3-channel JPEGs and were converted to single-channel. No label file is empty, malformed or out of bounds.

Near-duplicates. A difference hash over all templates gave 337 candidate pairs, re-checked at pixel level (best offset within ± 8 px, Jaccard ≥ 0.9 on copper and background). 40 duplicates were confirmed, all inside a single group, which supports group as the board unit. None crosses train/val; three test pairs (`50600065`, `92000114`, `92000118`) duplicate a train pair and are flagged so results can be reported with and without them.

Native misalignment. DeepPCB is described as aligned, but an integer search (± 8 px) for the offset that minimises template/inspected disagreement finds 123 pairs (8.2 %) where some offset cuts the XOR area by at least $1.5\times$: 75 train, 8 val, 40 test, typically 2–3 px and up to 8 px (Fig. 1). Their "aligned" condition is therefore already shifted. The offsets are stored per pair, so curves can be reported on all pairs and, as a sensitivity check, on the 1,377 clean pairs (460 in test). 17 templates are near-blank (< 2 % copper or background).

IV. CONTROLLED MISALIGNMENT PROTOCOL

The reference is never modified; the inspected image and its boxes are moved together. For each validation and test pair, ten conditions are generated (Table II): the pair as supplied (`s0`); integer shifts of 1, 2, 3 and 5 px along x or y, each with a random sign; and a *residual* condition r with a sub-pixel translation of 0.25–1.5 px in a random direction plus a rotation in $[-0.5°, 0.5°]$, which whole-pixel registration cannot undo exactly. Integer shifts copy pixels (nearest-neighbour), so they add no interpolation artefacts; r uses bilinear interpolation, and rotated boxes are replaced by the axis-aligned envelope of their four corners. Parameters are drawn from a generator seeded with $(6314, \text{pair ID})$, so both the baseline and the detector read the *same* files, independent of processing order. **Training does see misalignment:** the non-augmented run uses the 800 pairs as supplied (75 already natively offset), and the augmented run adds one random 1–5 px shift and one residual copy of every pair (2,400 images, separate seed stream; Section VI).

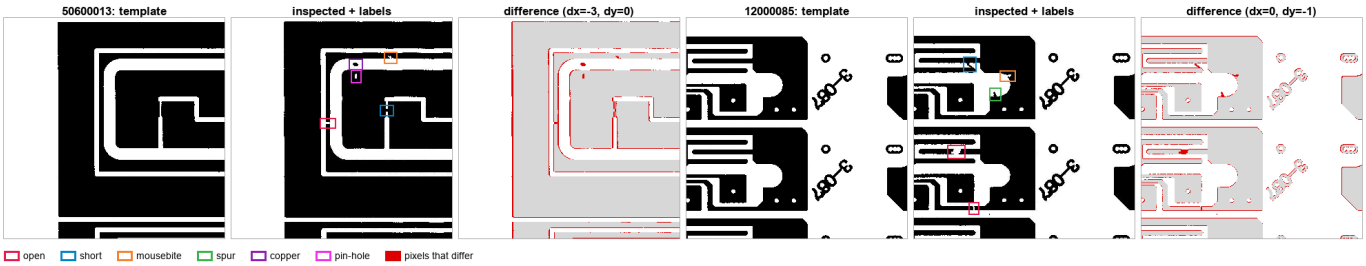


Fig. 1. Two pairs supplied as "aligned" but offset by 3 px and 1 px (offset found by the audit). Boxes show the labelled defects by class; red marks pixels that differ between template and inspected image, outlining every edge along the offset direction.

TABLE II
PREPARED INSPECTED IMAGES (ONE REFERENCE PER PAIR, 1,500 IN TOTAL).

Split	`s0`	`s1/2/3/5`×$\{x, y\}$	r	Images	Boxes
Train	800	–	–	800	5,338
Val	200	8×200	200	2,000	15,350
Test	500	8×500	500	5,000	31,400

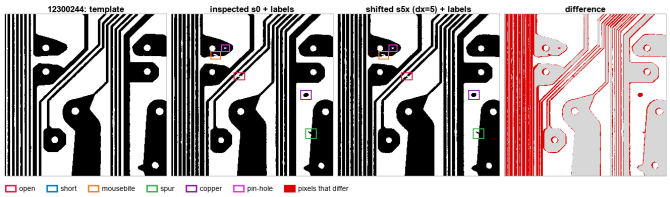


Fig. 2. Label check for the 5 px shift. Columns: template; inspected `s0` with labels; shifted image with shifted labels (boxes stay on their defects); difference in red, where the shift turns every track edge into a false difference.

Borders and truncated defects. Pixels uncovered by a shift are filled by edge replication. They lie inside an 8 px border (larger than the 5 px shift and the ≈4 px worst-case residual displacement) that is excluded at evaluation: predictions centred there are discarded. A box whose *original* position is within 14 px of an edge ($8 + 6$ px maximum displacement) is marked *ignore* in every condition, so the same defects are treated as "don't care" at every shift: 258 boxes (2.6 %). Ignored boxes count as neither hit nor miss, and a prediction matching one is not a false positive.

V. VERIFICATION

Two sets of checks were run: the pipeline's own QA scripts, and an independent audit written separately for this report that re-decodes the raw DeepPCB files with a different library (Pillow/SciPy instead of OpenCV) and does not reuse the pipeline code (Table III). The only difference found is one colour JPEG (`44000020`), where the two decoders' grey conversions differ on 2 % of pixels; its shifted copies still match its own `s0` exactly. Visual check sheets were also inspected for every condition (Fig. 2).

TABLE III
AUTOMATED CHECKS ON THE PREPARED DATA.

Check	Result
Pair lists, splits and `boxes.csv` match raw files	1500/1500
Reference images identical to raw templates	1500/1500
`s0` identical to raw inspected image (independent decoder)	1499/1500
Integer-shift image = `s0` shifted by (dx, dy), pixel-exact	5600/5600
Box count, class, (dx, dy) offset and ignore flag correct	52088/52088
Non-ignored boxes inside valid region $[8, 632]^2$	41220/41220
Residual images reproduced from stored parameters†	700/700
COCO files consistent with master CSV	21/21
YOLO images with matching labels; zip CRC intact	10200/10200

†Mean absolute difference ≤ 0.22 grey levels in the valid region; ECC re-registration of 120 samples gives a maximum corner error of 0.05 px.

VI. MODEL-READY FORMATS AND REPRODUCIBILITY

The master format is CSV: `splits.csv` (pair, group, split, native offset, duplicate flag), `images.csv` (one row per inspected image with dx, dy, rotation) and `labels.csv` (one row per box with class and ignore flag). The same data are exported as 21 COCO files (ignore → `iscrowd`) and as a YOLO package in which each pair is one 3-channel image (reference, inspected, |difference|), so an RGB-pretrained YOLOv10-Nano can be used unchanged. The augmented run (3× the images) trains for one third of the epochs, so both runs take the same number of steps. Colour augmentation stays off, since it would corrupt the difference channel; YOLO cannot encode ignore boxes, so evaluation reads `labels.csv`. Everything is rebuilt from the untouched DeepPCB download with `scripts/run_all.sh`.

VII. LIMITATIONS

Most defects are synthetic; validation is a single board group; each pair has only one residual variant; 123 pairs are natively misaligned; and 168 test pairs share a board with trainval. These are reported alongside every result rather than corrected.

REFERENCES

[1] S. Tang, F. He, X.-L. Huang, and J. Yang, "Online PCB defect detector on a new PCB defect dataset," *arXiv:1902.06197*, 2019.